

Poster Abstract: Studying the Security Threats of Partially Processed Deep Neural Inference Data in an IoT Device

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ABSTRACT

Partial computation offloading offers lower latency, privacy (by processing data close to the source), and higher performance. In partial offloading, some of the processing is performed on the less capable Internet-of-Things (IoT), and more complex computation is performed in more capable cloud computers. Though many well-established security protocols are available to address external intrusion, it introduces significant overhead for large data sample. This overhead is becoming a challenging problem with the recent trend and promises of on-device deep neural network (DNN) inference. This paper studies how much prior data distribution knowledge is required by an intruder to retrieve the input from a partially computed vector. We explore simple and complex DNN tasks, which are unknown to the intruder. We show that for simple datasets, Auto-Encoder(AE) and Variational Auto Encoder(VAE) can retrieve the original input image with a 4% accuracy drop on average. However, the same is not valid for complex data distribution.

CCS CONCEPTS

• Computer systems organization → Embedded software.

KEYWORDS

Mobile Computing, Inverse Deep Neural Network

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1 INTRODUCTION

With the widespread popularity of omnipresent edge devices, performance, data privacy, and quick response are necessary. These edge devices primarily belong to the low- or zero-power Internet of Things (IoT) family. Despite the growing capabilities of tiny edge devices (e.g., neural accelerators), they often fail to achieve accurate and timely responses with on-device computing only. Thus, they rely on cloud computers for additional processing due to their

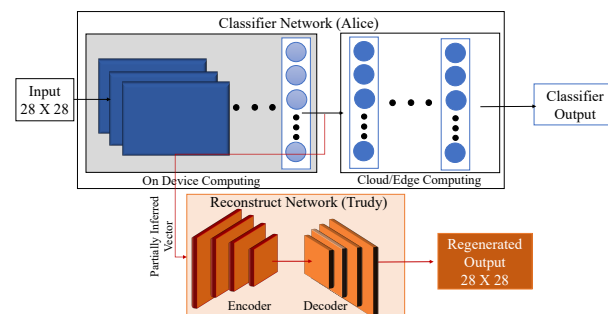


Figure 1: The reconstructor network retrieves the input from the partially inferred vector of the classifier

highly efficient processing and memory capabilities. However, privacy and security threats encourage processing the data close to the source of data. Besides, data transmission requires significant energy consumption for low-power edge devices [1]. Moreover, the networking overhead will be significant when we fulfill our prediction of having a trillion IoT devices by 2035. As a result, partial on-device computing shows promise by performing partial computation on-device with additional computation on the cloud to provide a more accurate result with lower latency.

Though the innovation of backscatter shows the possibility of transmitting with zero power, the encryption overhead of large data samples still exists. Low overhead encryption, e.g., Advanced Encryption Standards (AES), still accumulate significant overhead for large samples. Thus, extremely low-power IoTs are susceptible to external intrusion, e.g., input retrieval from processed vectors.

The authors wonder if the partially inferred DNN vector, which passes through several non-linear functions, inherently has the potential to preserve security. Literature on the adversarial attacks on DNN either determines the target process (DNN model) [4] or reconstructs input through the final layer's distribution and prior knowledge of data distribution [3]. The final layer of DNN contains the most significant features that assist the reconstruction.

This work focuses on the input retrieval from partially inferred vectors of the earlier layer of DNN that are yet to extract the significant features. We present a comprehensive study on how much prior information is required for an intruder to build an inverse DNN to retrieve the input from partially processed vectors. We implement classifiers of different complexity using fully connected (FC) neural networks and convolutional neural networks (CNN). We then implement two encoder-decoder models with AutoEncoder (AE) and Variational AutoEncoder (VAE) to retrieve the input from the partially calculated vectors. Furthermore, we present the effectiveness of the inverse DNN in terms of accuracy as a function of available prior information. Unlike Federated Learning where on-device computation is required to update the model, we only perform partial on-device inference due to constrained-resource.

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2 PROBLEM FORMULATION

Threat Model. Suppose Alice's IoT (e.g., MSP430) infers the first n layers of a DNN classifier and sends the output vector of the n th layer to cloud computing for further processing. In low-power communication, sending raw image data is 49x more energy-hungry than processing on-device [1]. However, as her IoT device's memory is insufficient for processing the whole data, she relies on the cloud for additional computation. The encryption before sending the vector introduces significant overhead (a 256-dimensional vector requires 21 W for encryption [2], impacting her time and energy budget. An intruder named Bob quietly sniffs the transmitted packets to retrieve Alice's input to the DNN. Let us assume that Bob has no knowledge of the DNN architecture Alice is inferring. However, he has a small dataset that reflects Alice's data distribution but was not used to train Alice's DNN. Bob can also send input to Alice's IoT device and sniff the transmission for partially inferred vectors.

Details of Classifier Network. We study two classifier networks – simple Fully Connected Network (FCN) and complex Convolutional Neural Network (CNN). The FCN takes a flattened input image (28X28) and passes it through two fully connected layers, each with 256 neurons. This dimension reduction increases the number of trainable parameters exponentially, making it difficult to invert the network despite the simple architecture. The CNN consists of 3 convolution (kernel size: 3x3) and four fully connected layers.

Details of Reconstruction Network. We consider auto-encoders as reconstruction networks to retrieve the input due to their regeneration capabilities. We explore Auto-Encoder (AE) and Variational Auto Encoder (VAE) to regenerate Alice's input by only providing a single hidden layer output from the classifier. AE/VAE models consist of an encoder, a latent vector, and a decoder. Despite similar regeneration quality and components, AE and VAE have different working principles. AE's latent vector has no specific distribution and reflects a form of deconvolution network, while VAE's latent vector is a distribution of the input space. Thus, VAE performs better than AE for unseen data due to its probability dependant principle. The encoder has three layers (128, 64, and 32 neurons), and the decoder has six layers (32, 64, 128, 256, 512, and 784 neurons). In VAE, the additional mean and the standard deviation are added with the original 20-dimensional latent vector. Input and output of the AE/VAE are 256 and 28x28 dimensional vectors, respectively.

3 OUR FINDINGS

Dataset. This study uses two popular image datasets – MNIST and CIFAR-10. First, we split the dataset into 80% training and 20% training samples. Next, we randomly remove $N\%$ samples from the training set, where $N = 5, 10, 15, 20, \text{ and } 30$. These $N\%$ samples are available to Bob and have not been used to train the classifiers to ensure that Bob has no prior knowledge. We explore the effect of this variable N on Bob's reconstruction capability.

Performance Metric. We denote the classifier accuracy with the original test dataset as AC. The reconstructed image infers the classifier and that AR is the resultant accuracy. We report accuracy drop, $AD = AC - AR$, as the performance metric. AD signifies how close the reconstructed image is to the original input image. A higher accuracy drop indicates failure in reconstruction.

Effect of N . For MNIST, Figure 2(a)-(b) shows that with a lower value of N , or in other words, the smaller available data reflecting

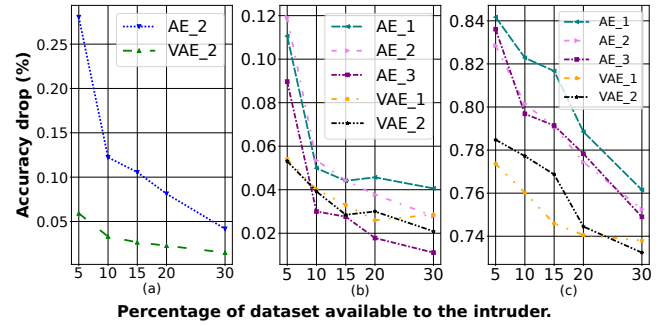


Figure 2: Performance of input retrieval for (a-b) MNIST FCN-CNN, and (c) CIFAR-10

the training data distribution Bob has, the higher accuracy drop he will experience. We also observe a prominent elbow near 10% for both simple and complex networks. Thus, we can conclude that with only 10% of data reflecting the data distribution but unseen to the classifier, Bob can reconstruct Alice's input with a median of 4% accuracy drop. For CIFAR-10, Figure 2(c) demonstrated a significant drop in accuracy even with a higher value of N , implying that AD is not only dependent on the value of N .

Effect of Network Architecture. In Figure 2, VAE outperforms AE for both MNIST and CIFAR data. Moreover, VAE does not experience as sharp a accuracy drop as AE with decreasing value of N . VAE performs better than AE due to the variability added by the mean and standard deviation, which results in generalization and better reconstruction of unseen data. Despite being a simpler network architecture of the classifier, data inferred through FCN is more challenging to retrieve than CNN. As previously mentioned, FCN has higher trainable variables contributing to further complexity despite the simpler architecture. Using VAE, a vector from the CNN can be retrieved only with 5% data available to Bob.

Effect of Inferred Depth. Figure 2 reveals that the more DNN layers are inferred on-device, the more accurate the reconstructed image is. As deeper layers on a DNN extracts more significant features and discard the redundant information, the vectors closer to the output layer will experience less accuracy drop.

Effect of Dataset Complexity. Finally, this study reveals that both AE and VAE suffers unacceptable accuracy drop for more complex and hard to classify dataset, e.g., CIFAR-10. As the input image goes through more successive operations that involve dropout and max-pooling, the resultant vector is harder to retrieve. In this scenario, only FCN or CNN is insufficient for input retrieval in a complex dataset. We will require a sophisticated reconstructor network to retrieve such information, which we will explore in future works.

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